

Digital Organizations: Sociotechnical Systems and the Algorithmic Enterprise

Executive Summary

Digital transformation reshapes organizations from purely human systems into **sociotechnical systems** — hybrid ecologies where human agents, software agents, algorithms, and data infrastructure form an integrated whole. Under Active Inference, digital organizations are systems where the Markov blanket is partially composed of technical infrastructure: APIs, databases, and algorithms that mediate the boundary between the organization and its environment. This module examines sociotechnical systems design, the human-technology interface, legacy system transformation, and the architecture of digitally native organizations.

Learning Objectives

1. Define **sociotechnical systems** and explain why digital transformation is fundamentally about system design, not technology adoption
 2. Apply the **Markov blanket concept** to understand how technology reshapes organizational boundaries
 3. Analyze the difference between **digitization** (converting analog to digital), **digitalization** (changing business processes), and **digital transformation** (changing the business model)
 4. Understand **technical debt** as accumulated model error in organizational infrastructure
 5. Design **human-technology interfaces** that leverage the complementary strengths of human and algorithmic agents
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Key Concepts

1. The Sociotechnical System

Component	Human System	Technical System	Integration Challenge
Sensing	Human perception, intuition	Sensors, data collection, APIs	Ensuring humans and machines sense complementary signals
Processing	Human judgment, creativity	Algorithms, ML models	Deciding which decisions are human vs. algorithmic
Acting	Human execution, relationship	Automation, robotics	Coordinating human and machine actions
Learning	Training, culture change	Model retraining, data update	Keeping human and machine knowledge in sync

2. Three Levels of Digital Change

Level	What Changes	Example	Organizational Impact
Digitization	Media format	Paper records → database	Minimal — same process, different medium
Digitalization	Business processes	Manual approval → automated workflow	Moderate — processes change, model stays
Digital transformation	Business model	Retail store → e-commerce platform	Fundamental — the generative model itself changes

3. Technology as Markov Blanket

Case Study — Amazon's Algorithmic Organization: Amazon's organizational boundary is largely technological. APIs connect internal services, recommendation algorithms mediate customer interactions, and warehouse robots execute physical logistics. The technology doesn't just support the organization — it constitutes the organizational boundary. This means organizational design and technology architecture are the same problem.

4. Technical Debt as Model Error

Technical debt accumulates when short-term implementation choices create long-term maintenance costs. Under Active Inference, technical debt is accumulated model error — the gap between what the system assumes and what it actually needs to do.

Cross-References

- For organizational systems, see [Organizational Systems: Systems](#)
 - For teams as systems, see [Collective Intelligence: Systems](#)
 - For competitive ecosystems, see [Strategic Modeling: Systems](#)
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Summary

Concept	Digital Transformation Meaning
System	Sociotechnical system — human + technology integrated whole
Markov blanket	APIs, data infrastructure, algorithms as organizational boundary
Digitization vs. digitalization vs. transformation	Three levels of digital change with increasing organizational impact
Technical debt	Accumulated model error in organizational technology
Human-technology interface	The integration point between human and algorithmic agents

References

- Trist, E. L. (1981). *The evolution of sociotechnical systems*. Toronto: Ontario Quality of Working Life Centre.
- Westerman, G. et al. (2014). *Leading Digital*. Harvard Business Review Press.